

SUMMARY

AI/ML Software Engineer spanning agentic AI, computer vision, and full-stack engineering — building both the models and the products around them. Currently launching an open-source platform for training, deploying, and sharing digital pathology models and pipelines; previously built fraud and compliance analytics in fintech.

EXPERIENCE

Epivara, Inc., BioTech

AI/ML Software Engineer

Champaign, IL

May 2025 – Present

- Built and pretrained DINOv3 ViT + CNN-adapter model capturing tissue-scale structure down to sub-nuclear detail: +18.5% mSA and +4.6% F1 over a state-of-the-art baseline on Leydig-cell histology; underpinned a toxicology study.
- Engineered a $16\times$ larger field of view at $2\times$ lower inference cost and $3.7\times$ faster training over a state-of-the-art baseline, supplying the anatomical context needed to reject biologically implausible classifications across $\sim 3/4$ of surveyed tissue endpoints.
- Creating a multi-view vision system that combines SAM 3.1, multi-animal 3D pose tracking, and temporal event modeling to automate behavior measurement under severe occlusion.
- Shipped an agentic data analysis platform with LLM tool-calling and MCP support that autonomously explores schemas, writes SQL, executes sandboxed Python, and generates reports.
- Developed a fault-tolerant human-in-the-loop data ingestion agent with guardrails that autonomously parses, validates, and structures experimental datasets, pausing for user approval – reducing insertion time from hours to minutes.

Tinkoff, FinTech

Data Analyst (Compliance)

Moscow, Russia

Feb 2022 – Oct 2022

- Contributed to feature development for a new Anti-Money Laundering model that increased fully automated task resolutions from 45% to 60% and cut annual costs by \$1.7M.
- Developed SQL/Python/Spark pipelines and dashboards that improved fraud and compliance analysis, cutting 1,000 hours of manual work annually.

Tinkoff, FinTech

Data Analyst (Insurance)

Moscow, Russia

Mar 2021 – Feb 2022

- Engineered a real-time quality measurement framework that merged backend logs, frontend events, and database tables, exposing live SRE metrics in Grafana and raising product quality (aggregated metric) by 18% while saving \$60K monthly.
- Collaborated with product, engineering, and support teams to diagnose product issues and run a quality improvement loop.
- Delivered ML-ready datasets for churn prediction and ran A/B tests to measure conversion impacts.

PROJECTS

HistoForge: Open-Source AI Platform for Digital Pathology (Launching Aug 2026)

[Link]

- Built an end-to-end platform (QuPath Java plugin + Modal GPU inference runtime + Hugging Face model hub + pip package) that brings foundation-model tissue analysis to pathologists and lets labs share fine-tuned models and pipelines.
- Built a large field-of-view DINOv3 + CNN-adapter model pretrained on 16 public cell-segmentation datasets; shipping 25 nuclei- and structure-level modules across oncology and toxicology, trained on public and proprietary data.
- Packaged training and inference so labs can fine-tune, run, and share their own models and pipelines, targeting reproducible quantitative pathology for oncology and toxicology research.

Entropy-aware sampling in vLLM

[Link]

- Implemented entropy-aware token sampling in vLLM with GPU-batched lookahead to control diversity by penalizing entropy-reducing tokens in speculative decoding.

EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science | GPA 3.8 / 4 | Graduate Research Assistant

Champaign, IL

Aug 2024 – May 2026

Bauman Moscow State Technical University

Specialist in Autonomous Informational and Control Systems | GPA 4.7 / 5

Moscow, Russia

Sep 2016 – Jun 2022

SKILLS

Languages: Python (FastAPI, SQLAlchemy, asyncio), SQL, TypeScript (React, Zustand, shadcn/ui), Java, C++, MATLAB

AI/ML: PyTorch, LangGraph, vLLM, Scikit-learn, W&B, tool and agent orchestration, HITL, RAG

Infra/Data: AWS (EC2, S3, IAM, ECS, VPC), Modal, Docker, PySpark, PostgreSQL, HPC (SLURM)